Guru Nanak Institute of Technology

BRANCH: Common to All Branches

LABORATORY MANUAL

COMPUTER PROGRAMMING LAB

CLASS: I B. Tech I & II Semester

FACULTY: - J.NARESH KUMAR

ASSISTANT PROFESSOR, IT DEPT.

COMPUTER PROGRAMMING IN C LAB SYLLABUS

B.Tech. I Year II Sem.
Course Code: **CS108ES/CS208ES 0 0/3/0 2**

Course Objective:
To write programs in C using structured programming approach to solve the problems.

Course Outcomes
1. Ability to design and test programs to solve mathematical and scientific problems.
2. Ability to write structured programs using control structures and functions.

Recommended Systems/Software Requirements:
- Intel based desktop PC
- GNU C Compiler

1. a) Write a C program to find the factorial of a positive integer.
b) Write a C program to find the roots of a quadratic equation.

2. a) Write a C program to determine if the given number is a prime number or not.
b) A Fibonacci sequence is defined as follows: the first and second terms in the sequence are 0 and 1. Subsequent terms are found by adding the preceding two terms in the sequence. Write a C program to generate the first n terms of the sequence.

3. a) Write a C program to construct a pyramid of numbers.
b) Write a C program to calculate the following sum

$Sum=1- x^2/2! + x^4/4! - x^6/6! + x^8/8! - x^{10}/10!$

4. a) The least common multiple (lcm) of two positive integers a and b is the smallest integer that is evenly divisible by both a and b. Write a C program that reads two integers and calls lcm (a,b) function that takes two integer arguments and returns their lcm. The lcm function should calculate the least common multiple by calling the gcd (a,b) function and using the following relation :
LCM (a, b)=a b/ gcd (a, b)

b) Write a C program that reads two integers n and r to compute the ncr value using the following relation: ncr (n,r) = n! / r! (n-r)! . Use a function for computing the factorial value of an integer.

5. a) Write C program that reads two integers x and n and calls a recursive function to compute x^n
b) Write a C program that uses a recursive function to solve the Towers of Hanoi problem.
c) Write a C program that reads two integers and calls a recursive function to compute nc_r value.

6. a) Write a C program to generate all the prime numbers between 1 and n, where n is a value supplied by the user using Sieve of Eratosthenes algorithm.

b) Write a C program that uses non recursive function to search for a Key value in a given list of integers. Use linear search method.

7. a) Write a menu-driven C program that allows a user to enter n numbers and then choose between finding the smallest, largest, sum, or average. The menu and all the choices are to be functions. Use a switch statement to determine what action to take. Display an error message if an invalid choice is entered.

b) Write a C program that uses non recursive function to search for a Key value in a given sorted list of integers. Use binary search method.

8 a) Write a C program that implements the Bubble sort method to sort a given list of integers in ascending order.

b) Write a C program that reads two matrices and uses functions to perform the following:
i) Addition of two matrices
ii) Multiplication of two matrices

9. a) Write a C program that uses functions to perform the following operations:
i) to insert a sub-string into a given main string from a given position.
ii) to delete n characters from a given position in a given string.

b) Write a C program that uses a non recursive function to determine if the given string is a palindrome or not.

10. a) Write a C program to replace a substring with another in a given line of text.
b) Write a C program that reads 15 names each of up to 30 characters, stores them in an array, and uses an array of pointers to display them in ascending (ie. alphabetical) order.

11. a) 2's complement of a number is obtained by scanning it from right to left and complementing all the bits after the first appearance of a 1. Thus 2's Complement of 11100 is 00100. Write a C program to find the 2's complement of a binary number.
b) Write a C program to convert a positive integer to a roman numeral. Ex. 11 is Converted to XI.

12. a) Write a C program to display the contents of a file to standard output device.
b) Write a C program which copies one file to another, replacing all lowercase characters with their uppercase equivalents.

13. a) Write a C program to count the number of times a character occurs in a text file. The file name and the character are supplied as command-line arguments.

b) Write a C program to compare two files, printing the first line where they differ.

14. a) Write a C program to change the nth character (byte) in a text file. Use fseek function.

b) Write a C program to reverse the first n characters in a file. The file name and n are specified on the command line. Use fseek function.

15. a) Write a C program to merge two files into a third file (i.e., the contents of the first file followed by those of the second are put in the third file).

b) Define a macro that finds the maximum of two numbers. Write a C program that uses the macro and prints the maximum of two numbers.

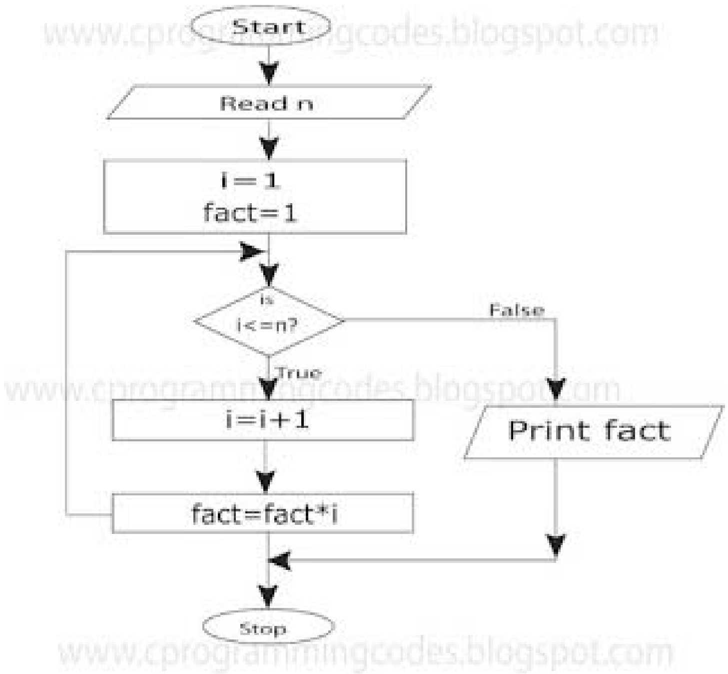
1. a) Write a C program to find the factorial of a positive integer.

Description:

The user will supply an integer value and for that value need to calculate the factorial without using recursion. Let value=4 then 4! =4*3*2*1=24

Algorithm:

step 1. Start
step 2. Read the number n
step 3. [Initialize]
 i=1, fact=1
step 4. Repeat step 4 through 6 until i=n
step 5. fact=fact*i
step 6. i=i+1
step 7. Print fact
step 8. Stop



Program:

```
#include<stdio.h>
#include<conio.h>
void main()
{
 int n,i,fact=1;
 printf("Enter any number : ");
 scanf("%d", &n);
 for(i=1; i<=n; i++)
    fact = fact * i;
 printf("Factorial value of %d = %d", n, fact);
 getch();
}
```

Output:

enter any integer 5
the factorial of a given number =120

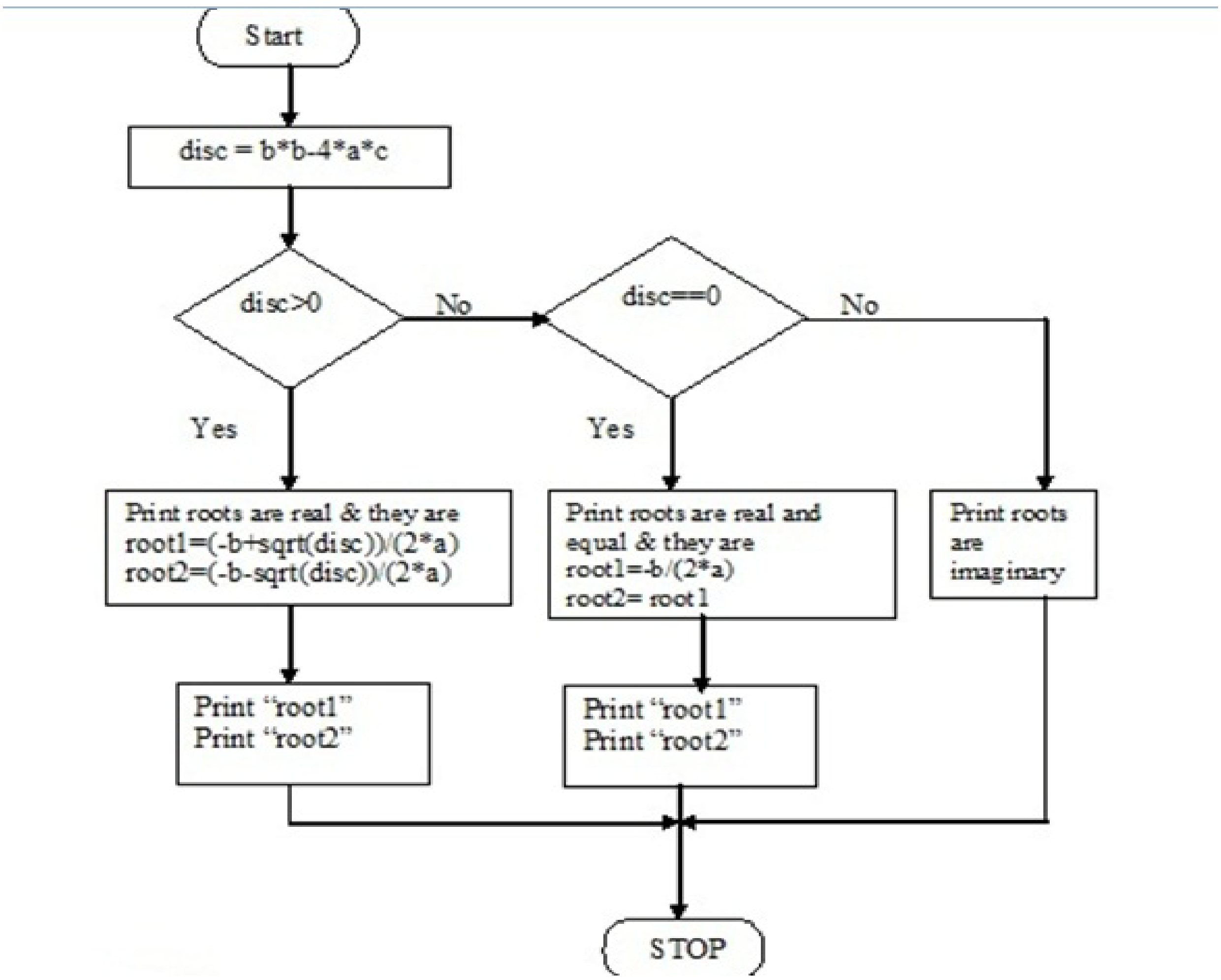
1.b) Write a C program to find the roots of a quadratic equation.

Description:

Roots of quadratic equation are $\frac{-b\pm\sqrt{b^2-4ac}}{2a}$

Algorithm:

Step 1: Start
Step 2: Read a,b,c
Step 3: calculate disc = b*b-4*a*c
Step 4: if(disc>0)
Begin
Step 5: root1=(-b+sqrt(disc))/(2*a)
Step 6: root2=(-b-sqrt(disc))/(2*a)
Step 7: Print "Root1" , "Root2"
End
Step 8: else if(disc=0)
Begin
Step 9: root1=-b/(2*a)
Step 10: root2=root1;
Step 11: Print "Root1" , "Root2"
End
Step 12: else
Step 13: Print Roots are imaginary
Step 14: Stop



Program:

```
#include<stdio.h>
#include<conio.h>
#include<math.h>
void main()
{
int a,b,c;
float d,r1,r2;
clrscr();
printf("enter values of a,b &c");
scanf("%d%d%d",&a,&b,&c);
d=b*b - 4*a*c;
if( d==0)
{
r1=r2=-b/2*a;
printf("roots are equal\n");
printf("root1=root2=%f",r1);
}
else if(d>0)
{
r1=-b+sqrt(d)/(float)2*a;
r2=-b-sqrt(d)/(float)2*a;
printf("roots are real\n");
printf("root1=%f\troot2=%f",r1,r2);
}
else
{
printf("roots are imaginary");
}
getch();
}
```

Output1:
enter values of a,b &c 2 4 2
roots are equal
root1=root2=-4.000000

Output2:
enter values of a,b &c 1 4 2
roots are real
root1=-2.585786 root2=-5.414214

Output3:
enter values of a,b &c 1 4 5
roots are imaginary

2. a) Write a C program to determine if the given number is a prime number or not.

Program :

```
#include<stdio.h>
#include<conio.h>
void main()
{
int i,n,count=0;
clrscr();
printf("enter any number");
scanf("%d", &n);

for(i=0; i<=n; i++)
  {
   if(n%i==0)
   count++;
  }
if(count==2)
printf("Entered number  %d is a prime number, n);
else
printf("Entered number  %d is not a prime number, n);
getch();
}
```

Output:

enter any number 20
Entered number 20 is not a prime number

enter any number 19
Entered number 19 is a prime number

2. b) A Fibonacci sequence is defined as follows: the first and second terms in the sequence are 0 and 1. Subsequent terms are found by adding the preceding two terms in the sequence. Write a C program to generate the first n terms of the sequence.

Description:

Initial Fibonacci numbers are 0 and 1. Next number can be generated by adding two numbers. So 0+1=1. Therefore next number can be generated by adding two previous . so Fibonacci series is 0 1 1 2 3 5 ……

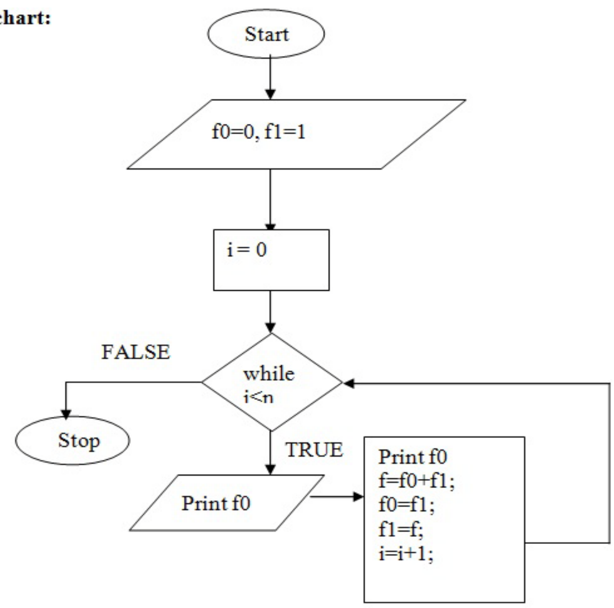
Algorithm:

```
Step 1  : Start
Step 2  : Read n
Step 3  : Initialize f0 ← 0, f1 ← 1, f ← 0
Step 4  : i=0
Step 5 : while(i<=n) do as follows
            printf("%d\t",f0);
         f=f0+f1;
         f0=f1;
         f1=f;
         i=i+1;
           If not goto step 7
Step 6  : Stop
```

Program :

```
#include<stdio.h>
#include<conio.h>
void main()
{
int i,n,a=0,b=1,c;
clrscr();
printf("enter any number");
scanf("%d",&n);
printf(" the fibonacci sequence");
printf("%d\t%d\t",a,b);
for(i=3; i<=n;i++)
{
c=a+b;
printf("%d\t",c);
a=b;
b=c;
}
getch();
}
```

Flowchart:



Output:

enter any number 8
the fibonacci sequence0 1 1 2 3 5 8 13

3. a) Write a C program to construct a Pyramid of numbers.

Description: The user will supply the length of the pyramid then the numbers will be printed in the pyramid form.
Ex: length=3

```
        1
     2     3
  4     5     6
```

Algorithm:
Step 1: Start
Step 2: Read n
Step 3: i:=0
Step 4: if(i<n) goto step 5
        else goto step 15
Step 5: k=n
Step 6: if(k>i) goto step 7
        else goto step 10
step 7:print blank
step 8:k:=k-1
step 9:goto step 6
step 10:j=1;
step 11:if(<=i) goto step 12
        else goto step 15
step12:print j
step 13:j:=j+1
step 14: goto step 11
step 15:stop

Program:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n,i,j,k,x=1;
clrscr();
printf("Enter no of rows");
scanf("%d",&n);
for(i=1;i<=n;i++)
{
for(k=n-i;k>=1;k--)
printf(" ");
for(j=1;j<=i;j++)
{
printf("%4d",x);
x++;
}
printf("\n");
}
getch();
}
```

Output: **Enter no of rows 4**

```
    1
   2 3
  4 5 6
 7 8 9 10
```

3. b) Write a C program to calculate the Sum=1- $x^2/2! + x^4/4! - x^6/6! + x^8/8! - x^{10}/10!$

Description:

Ex: Let x=2

Sum=1- $x^2/2! + x^4/4! - x^6/6! + x^8/8! - x^{10}/10!$

Algorithm:

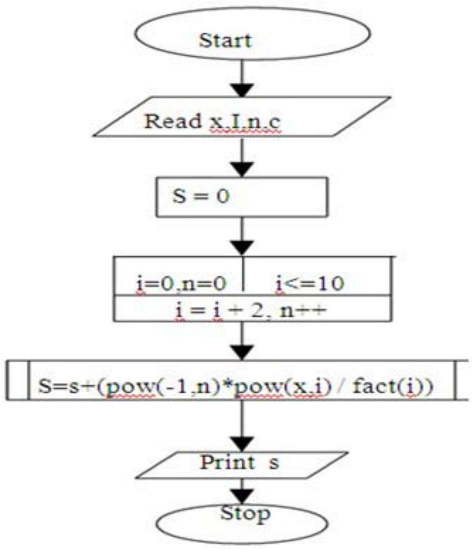
Main program:

Step 1: start
Step 2: declare x,i,n,s=0,c
Step 3: read x value
Step 4: for i=0 , n=0; i<=10; i=i+2, n++ goto step 5
Step 5: s=s+(pow(-1,n)*pow(x,i)/fact(i))
Step 6: print s value
Step 7: stop

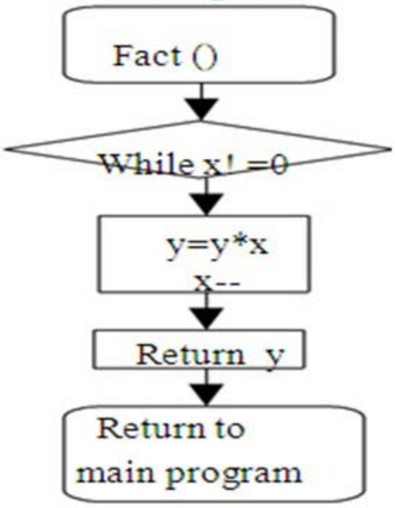
Sub program:_ fact(x)

Step 1: while x!=0 goto Step 2
Step 2: y=y+x; x—
Step 3: return y
Step 4: return to main program

Flowchart: **Main Program:**



Sub Program



Program:

```
#include<stdio.h>
#include<math.h>
long fact(int);

void main()
{
int x,i,n;
float s=0,c;
clrscr();
printf("\n enter the value of x \t");
scanf("%d",&x);     /*perform the looping operation*/
for(i=0,n=0;i<=10;i=i+2,n++)
s=s+(pow(-1,n)*pow(x,i)/fact(i));
printf("\n the result is %f",s);
getch();
}

long fact(int x)       /* calling sub program*/
{
int i;
long f=1;
for(i=1;i<=x;i++)
{
f=f*i;
}
return f;
}
```

Output:

Enter the value of x : 1
The result is 0.540302

4. a) The least common multiple (lcm) of two positive integers a and b is the smallest integer that is evenly divisible by both a and b. Write a C program that reads two integers and calls lcm (a, b) function that takes two integer arguments and returns their lcm. The lcm (a, b) function should calculate the least common multiple by calling the gcd (a, b) function and using the following relation: LCM (a,b) = a*b / gcd (a,b)

Program :

```
#include <stdio.h>
#include<conio.h>

/* Another way to write GCD Program
int gcd(int a, int b)    // Recursive function to return gcd of a and b
{
   if (a == b)
       return a;          // a is greater

   if (a > b)
       return gcd(a-b, b);
   return gcd(a, b-a);
}
*/

int lcm(int a, int b)          // Function to return LCM of two numbers
{
   return (a*b)/gcd(a, b);
}

int gcd(int a, int b)
{
   if(b == 0)
       return a;
   else
       return gcd(b, a%b);
}

void main()  // Driver program to test above function
{
   int a = 15, b = 20;
   clrscr();
   printf("LCM of %d and %d is %d ", a, b, lcm(a, b));
   getch();
}
```

Output:

LCM of 15 and 20 is 60

4. b) Write a C program that reads two integers n and r to compute the ncr value using the following relation: ncr(n,r)=n!/r!(n-r)!. Use a function for computing the factorial value of an integer.

Program:

```
#include<stdio.h>
#include<conio.h>

long fact(int);
long ncr(int, int);
long npr(int, int);

void main()
{
   int n, r;
   long ncr_result, npr_result;
   clrscr();
   printf("Enter the value of n and r\n");
   scanf("%d%d",&n,&r);
   ncr_result = ncr(n, r);
   npr_result = npr(n, r);
   printf("%dC%d = %ld\n", n, r, ncr_result);
   printf("%dP%d = %ld\n", n, r, npr_result);
   getch();
}

long ncr(int n, int r)
{
   long result;
   result = fact(n)/(fact(r)*fact(n-r));
   return result;
}

long npr(int n, int r)
{
   long result;
   result = fact(n)/fact(n-r);
   return result;
}

long fact(int n)
{
   int i;
   long f = 1;
   for (i = 1; i <= n; i++)
      f = f*i;
   return f;
  }
```

Output:

Enter the value of n and r 5 2
$5C_2$ = 10
$5P_2$ = 20

5. a) Write a C program that reads two integers X and N and calls a recursive function to compute X power N

Program:

```
#incude<stdio.h>
#include<conio.h>

int xpown(int,int);

void main()
{
int X,N,Result;
clrscr();
printf("Enter the values of X and N");
scanf("%d",&X,&N);
Result=xpown(X,N);
printf("The result is %d",  result);
getch();
}

int xpown(int X, int N)
{
if(N==0)
return 1;
else
return X*xpown(X,N-1);
}

}
```

Output:
Enter the values of X and N : 5   3
The result is 125

5. b) Write a C program that uses a recursive function to solve the Towers of Hanoi problem.

```
#include <stdio.h>          // C recursive function to solve tower of hanoi puzzle
void TowerOfHanoi(int n, char source, char destination, char auxiliary)
{
   if (n == 1)
   {
      printf("\n Move disk 1 from tower %c to tower %c", source, destination);
      return;
   }
   TowerOfHanoi(n-1, source, auxiliary, destination);
   printf("\n Move disk %d from tower %c to tower %c", n, source, destination);
   TowerOfHanoi(n-1, auxiliary, destination, source);
}
```

```
int main()
{
   int n = 4;          // Number of disks
   TowerOfHanoi(n, 'A', 'C', 'B');        // A, B and C are names of towers
   return 0;
}
```

Output:

```
 Move disk 1 from tower A to tower B
 Move disk 2 from tower A to tower C
 Move disk 1 from tower B to tower C
 Move disk 3 from tower A to tower B
 Move disk 1 from tower C to tower A
 Move disk 2 from tower C to tower B
 Move disk 1 from tower A to tower B
 Move disk 4 from tower A to tower C
 Move disk 1 from tower B to tower C
 Move disk 2 from tower B to tower A
 Move disk 1 from tower C to tower A
 Move disk 3 from tower B to tower C
 Move disk 1 from tower A to tower B
 Move disk 2 from tower A to tower C
 Move disk 1 from tower B to tower C
```

For n disks, total 2n – 1 moves are required.

5. c) Write a C program that reads two integers and calls a recursive function to compute ncr value.

```
#include<stdio.h>
#include<conio.h>
int rec_ncr(int ,int );

void main()
{
int n,r;
printf("Enter n and r");
scanf("%d%d",&n,&r);
printf("The value of %dC%d is %d",n,r,rec_ncr(n,r));
getch();
}

int rec_ncr(int n,int r)
{
if(r==0|r==n)
  return 1;
else
  return rec_ncr(n-1,r-1)+rec_ncr(n-1,r);
}
```

Output:

Enter the value of n : 6
Enter the value of r : 2
The value of C(6,2) is : 15

6. a) Write a C program to generate all the prime numbers between 1 and n, where n is a value supplied by the user using Sieve of Eratosthenes algorithm.

Sieve of Eratosthenes------A prime number
is a natural number that has exactly two distinct natural number divisors: 1 and itself.

To find all the prime numbers less than or equal to a given integer n by Eratosthenes' method:

Create a list of consecutive integers from 2 through n: (2, 3, 4, ..., n).
Initially, let p equal 2, the smallest prime number.
Enumerate the multiples of p by counting to n from 2p in increments of p, and mark them in the list (these will be 2p, 3p, 4p, ... ; the p itself should not be marked).
Find the first number greater than p in the list that is not marked. If there was no such number, stop. Otherwise, let p now equal this new number (which is the next prime), and repeat from step 3.
When the algorithm terminates, the numbers remaining not marked in the list are all the primes below n.

To find all the prime numbers less than or equal to 30, proceed as follows.

First generate a list of integers from 2 to 30:
2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30.

The first number in the list is 2; cross out every 2nd number in the list after 2 by counting up from 2 in increments of 2 (these will be all the multiples of 2 in the list):
2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30.

The next number in the list after 2 is 3; cross out every 3rd number in the list after 3 by counting up from 3 in increments of 3 (these will be all the multiples of 3 in the list):
2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30.

The next number not yet crossed out in the list after 3 is 5; cross out every 5th number in the list after 5 by counting up from 5 in increments of 5 (i.e. all the multiples of 5):
2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30.

The next number not yet crossed out in the list after 5 is 7; the next step would be to cross out every 7th number in the list after 7, but they all have been crossed out at this point, as these numbers (14, 21, 28) are also multiples of smaller primes because 7*7 is greater than 30. The numbers not crossed out in the list at this point are all the prime numbers below 30:
2 3 5 7 11 13 17 19 23 29.

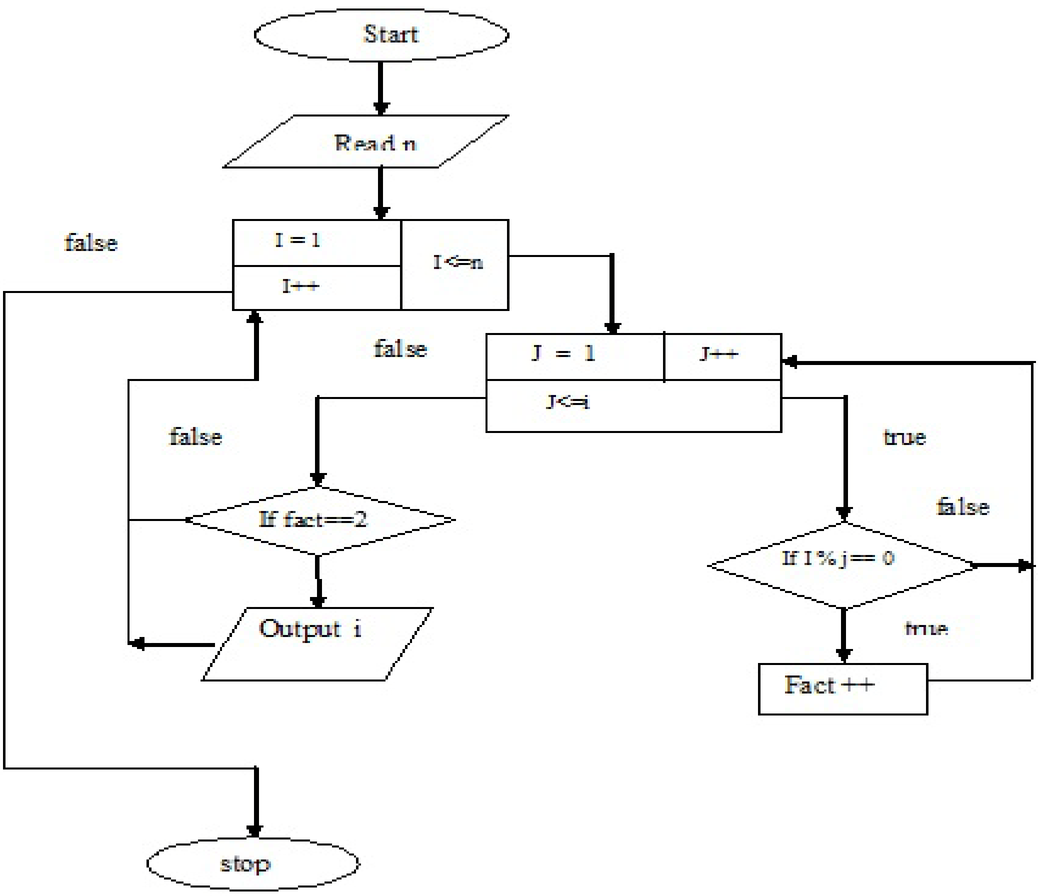
Description:

Prime number is a number which is exactly divisible by one and itself only
Ex: 2, 3,5,7,………;

Algorithm:

Step 1: start
Step 2: read n
Step 3: initialize i=1,c=0
Step 4:if i<=n goto step 5
If not goto step 10
Step 5: initialize j=1
Step 6: if j<=i do the following. If no goto step 7
i)if i%j==0 increment c
ii) increment j
iii) goto Step 6
Step 7: if c== 2 print i
Step 8: increment i
Step 9: goto step 4
Step 10: stop

Flowchart:



Program:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int i,j,n,count;
clrscr();
printf("Enter any number");
scanf("%d",&n);
printf("Prime numbers between 1 and %d\n", n);
```

```
for(i=1; i<=n; i++ )
{
count=0;
for(j=1; j<=i ;j++)
{
if(i%j==0)
count++;
}
if(count==2)
printf("%d\t",i);
}
}
```

Output:

Enter any number 20

Prime numbers between 1 and 20 : 2 3 5 7 11 13 17 19

6. b) Write a C program that uses non recursive function to search for a Key value in a given list of integers. Use linear search method.

Program:

```
#include<stdio.h>
#include<conio.h>
#include<process.h>

void main()
{
int i,n,element,a[10];
clrscr();
printf("enter how many elements u want to enter");
scanf("%d",&n);
printf("enter %d elements",n);
for(i=0;i<n;i++)
scanf("%d",&a[i]);
printf("enter any element to search");
scanf("%d",&element);
for(i=0;i<n;i++)
{
if(a[i]==element)
{
printf("element is found at position %d",i);
getch();
exit(0);
}
}
printf("element is not found");
getch();
}
```

Output:

enter how many elements u want to enter 5

enter 5 elements 20 30 10 40 50

enter any element to search 30

element is found at 1 position

7. a) Write a menu-driven C program that allows a user to enter n numbers and then choose between finding the smallest, largest, sum, or average. The menu and all the choices are to be functions. Use a switch statement to determine what action to take. Display an error message if an invalid choice is entered.

Program:

```
#include<stdio.h>
#include<conio.h>
#include<stdlib.h>

int largest(int a[10]);
int smallest(int a[10]);
int sum(int a[10]);
int average(int a[10]);

void main()
{
 int a[10],n,i,s,l,t,a1;
 clrscr();
printf("Enter array of 10 Numbers");
for(i=0;i<10;i++)
scanf("%d",&a[i]);
do
{
printf("\n  Menu :\n 1. Smallest of all the numbers.\n 2. Largest of all the numbers:");
printf("\n 3. Sum of all the numbers:\n 4. Average of all the numbers.\n 5.Exit");
printf("\n Enter your choice:");
scanf("%d",&n);
 switch(n)
 {
  case 1:
   s=smallest(a);
   printf("Smallest Number is %d",s);
   break;
 case 2:
   l=largest(a);
   printf("Largest Number is %d",l);
   break;
  case 3:
   t=sum(a);
   printf("Sum of all the Numbers is %d",t);
   break;
  case 4:
   a1=average(a);
   printf("Average of all the Numbers is %d",a1);
   break;
   case 5:
   exit(0);
  default:
   printf("Invalid Entry");
   break;
 }
printf("\n Do you want to continue?  y/n");
}while(getch()=='y');
```

```
getch();
}
int smallest(int a[10])
{
   int i,small=a[0];
   for(i=1;i<10;i++)
   if(a[i]<small)
     {
      small=a[i];
     }
return small;
}
int largest(int a[10])
{
   int i,large=a[0];
   for(i=1;i<10;i++)
   if(a[i]>large)
    {
     large=a[i];
    }
return large;
}
int sum(int a[10])
{
   int i,sum1=0;
   for(i=0;i<10;i++)
    sum1=sum1+a[i];
return sum1;
}
int average(int a[10])
{
  int i,avg,sum1=0;
  for(i=0;i<10;i++)
    sum1=sum1+a[i];
    avg=sum1/10;
return avg;
}
```

Output :
Enter array of 10 numbers 1 2 3 4 5 6 7 8 9 10

Menu :
1. Smallest of all the numbers.
2. Largest of all the numbers:
3. Sum of all the numbers:
4. Average of all the numbers.
5. Exit

Enter your choice: 1
Smallest Number is 1

Do you want to continue? y/n n

7. b) Write a C program that uses non recursive function to search for a Key value in a given sorted list of integers. Use binary search method.

Program :

```
#include<stdio.h>
#include<conio.h>
#include<process.h>
void main()
{
int i,n,element,a[10],low,high,mid;
clrscr();
printf("enter how many elements u want to enter");
scanf("%d",&n);
printf("enter %d elements",n);
for(i=0;i<n;i++)
scanf("%d",&a[i]);
printf("enter any element to search");
scanf("%d",&element);
low=0;
high=n-1;
while(low<=high)
{
mid = (low+high)/2;
if(element< a[mid])
high=mid-1;
else if (element > a[mid])
low=mid+1;
else
{
printf("element is found at position =%d",mid);
getch();
exit(0);
}
}
printf("element is not found");
getch();
}
```

Output:

```
enter how many elements u want to enter      5
enter 5 elements 10    20    30    40    50
enter any element to search         30
element is found at 1 position
```

8 a) Write a C program that implements the Bubble sort method to sort a given list of integers in ascending order.

```
#include<stdio.h>
#include<conio.h>

void main()
{
int i,j,n,a[100],temp;
clrscr();
printf("enter how many elements u want to enter");
scanf("%d",&n);
printf("enter %d elements",n);
for(i=0;i<n;i++)
scanf("%d",&a[i]);
for(i=0; i<n; i++)
{
for(j=i+1;j<n; j++)
{
if(a[i]>a[j])
{
temp=a[i];
a[i]=a[j];
a[j]=temp;
}
}
}
printf("sorted elements in ascending order\n");
for(i=0;i<n;i++)
printf("%d\n",a[i]);
getch();
}
```

Output:

enter how many elements u want to enter 5
enter 5 elements 20 30 10 50 40
sorted elements in ascending order 10 20 30 40 50

8. b) Write a C program that reads two matrices and uses functions to perform the following:
i) Addition of two matrices
ii) Multiplication of two matrices

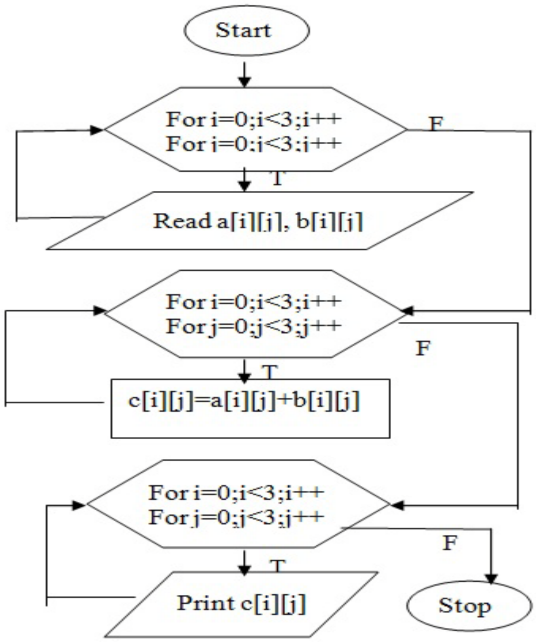
i) Addition of Matrices

Description:

The sum A+B of two m-by-n matrices A and B is calculated entry wise:

$(A + B)_{i,j} = A_{i,j} + B_{i,j},$

where $1 \le i \le m$ and $1 \le j \le n$.



Algorithm:

Step 1: Start
Step2: for i is 0 to 2 by step 1
for j is 0 to 2 by step 1
Step 3: Read a[i][j],b[i][j]
Step 4: goto step 2
Step 5: calculate c[i][j]=a[i][j]+b[i][j]
Step 6: goto step 2
Step 7: Print c[i][j]
Step 8: Stop

Program:

```
#include<stdio.h>
#include<conio.h>

void main()
{
int a[2][2],b[2][2],c[2][2],r1,c1,r2,c2,i,j;
clrscr();
printf("enter no of rows & columns of matrix-a");
scanf("%d%d",&r1,&c1);
printf("enter no of rows & columns of matrix-b");
scanf("%d%d",&r2,&c2);
if(r1==r2 && c1==c2)
{
printf("enter elements of matrix-a");
for(i=0;i<r1;i++)
{
for(j=0;j<c1;j++)
{
scanf("%d",&a[i][j]);
}
}
printf("enter elements of matrix-b");
for(i=0;i<r2;i++)
{
for(j=0;j<c2;j++)
{
scanf("%d",&b[i][j]);
}
}
printf("matrix addition:\n");
```

```
for(i=0;i<r1;i++)
{
for(j=0;j<c1;j++)
{
c[i][j]=a[i][j]+b[i][j];
printf("%2d",c[i][j]);
}
printf("\n");
}
}
else
printf("matrix addition is not possible");
getch();
}
```

Output :

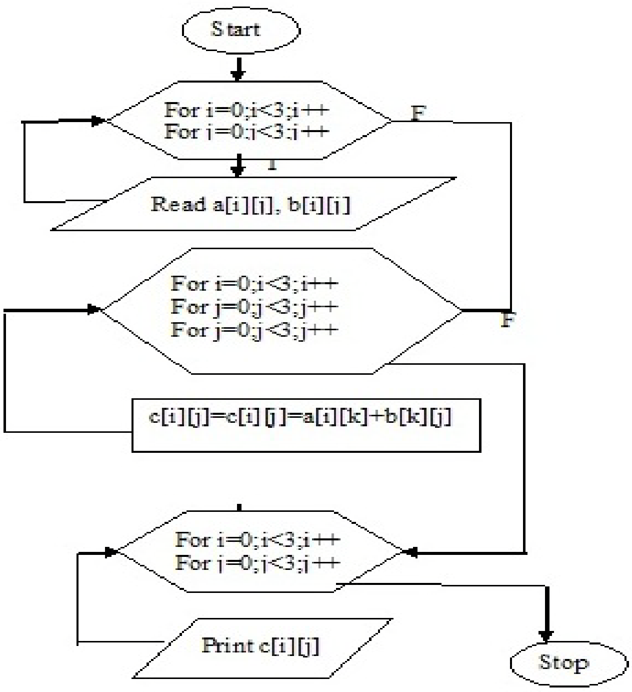
```
enter no of rows & columns of matrix-a  2       2
enter no of rows & columns of matrix-b  2       2
enter elements of matrix-a          1   2
                                    3   4

enter elements of matrix-b          5   6
                                    7   8
matrix addition:
                                    6    8
                                    10  12
```

ii) Multiplication Of Matrices

Description:

Multiplication of two matrices is defined if and only if the number of columns of the left matrix is the same as the number of rows of the right matrix. If **A** is an m-by-n matrix and **B** is an n-by-p matrix, then their matrix product **AB** is the m-by-p matrix.



Algorithm:

Step 1: Start
Step 2: for i is 0 to 2 by step 1
          for j is 0 to 2 by step 1
Step 3: Read a[i][j],b[i][j]
Step 4: goto step 2
Step 5: calculate c[i][j]=a[i][j]+b[i][j]
Step 6: goto step 2
Step 7: Print c[i][j]
Step 8: Stop

Program:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[2][2],b[2][2],c[2][2],i,j,m,n,p,q,k;
clrscr();
printf("\n enter the order of matrix-a");
```

```
scanf("%d%d",&m,&n);
printf("\n enter order of matrix b");
scanf("%d%d",&p,&q);
if(n==p)
{
printf("enter elements of matrix-a");
for(i=0;i<m;i++)
{
for(j=0;j<n;j++)
scanf("%d",&a[i][j]);
}
printf("enter elements of matrix b");
for(i=0;i<p;i++)
{
for(j=0;j<q;j++)
scanf("%d",&b[i][j]);
}
for(i=0;i<m;i++)
{
for(j=0;j<q;j++)
{
c[i][j]=0;
for(k=0;k<p;k++)
{
c[i][j]=c[i][j]+(a[i][k]*b[k][j]);
}
}
}
printf("\n multiplication of matrix is\n");
for(i=0;i<m;i++)
{
for(j=0;j<q;j++)
{
printf("%2d",c[i][j]);
}
printf("\n");
}
}
else
printf("\n matrix multiplication is not possible");
getch();
}
```

Output:

```
enter the order of matrix-a   2   2
enter order of matrix b       2   2

enter elements of matrix-a    1   2
                              3   4

enter elements of matrix b    1   1
                              1   1

multiplication of matrix is   3   3
                              7   7
```

9. a) Write a C program that uses functions to perform the following operations:
i) to insert a sub-string into a given main string from a given position.

Description:

The user will supply two strings(main string and sub string) and also the position where the second string will be inserted will be given by the user.

Ex: main string= comter substring=pu position to insert sub string=3

Result=computer

Program:

```
#include<stdio.h>
#include<conio.h>
#include<string.h>

void main()
{
   char str1[20], str2[20];
   int l1, l2, n, i;
   clrscr();
   puts("Enter the main string:\n");
   gets(str1);
   l1 = strlen(str1);
   puts("Enter the sub string:\n");
   gets(str2);
   l2 = strlen(str2);
printf("Enter the position in the main string where the sub string is to be inserted:\n");
   scanf("%d", &n);
   for(i = n; i < l1; i++)
    {
      str1[i + l2] = str1[i];
    }

  for(i = 0; i < l2; i++)
   {
      str1[n + i] = str2[i];
   }
      str2[l2 + 1] = '\0';
   printf("After inserting the sub string, the main string is %s:", str1);
   getch();
}
```

Output :

Enter Main String: program123
Enter Sub String: ming
Enter the position in the main string where the sub string is to be inserted: 7
After inserting the sub string, the main string is: Programming123

9. a) ii) to delete n characters from a given position in a given string.

Description:

The user will supply string and the position from where to start deletion and the number of characters to be deleted.

Ex: string= computer position to start deletion=2 position to insert sub string=2

Result=couter

Algorithm:

Step 1: start
Step 2: read string
Step 3: find the length of the string
Step 4: read the value of number of characters to be deleted and positioned
Step 5: string copy part of string from position to end, and
(position + number of characters to end)
Step 6: stop

Program :

```
#include<stdio.h>
#include<conio.h>
void main()
{
int i,pos,n;
char mstr[100];
void strdel();
clrscr();
printf("Enter the main string");
gets(mstr);
printf("\n Enter how many characters you want to delete and from which position you want to delete in the main string");
scanf("%d%d",&n,&pos);
strdel(mstr,n,pos);
getch();
}
void strdel(char m[],int n,int p)
{
int i;
for(i=p; m[i]!='\0';i++)
m[i]=m[i+n];
printf("After deletion the main string is: %s", m);
}
```

Output:

Enter the main string : hyderabad
Enter how many characters you want to delete and from which position you want to delete in the main string 3 3
After deletion the main string is: hydbad

9. b) Write a C program that uses a non recursive function to determine if the given string is a palindrome or not.

Description:

Palindrome means string on reversal should be same as original

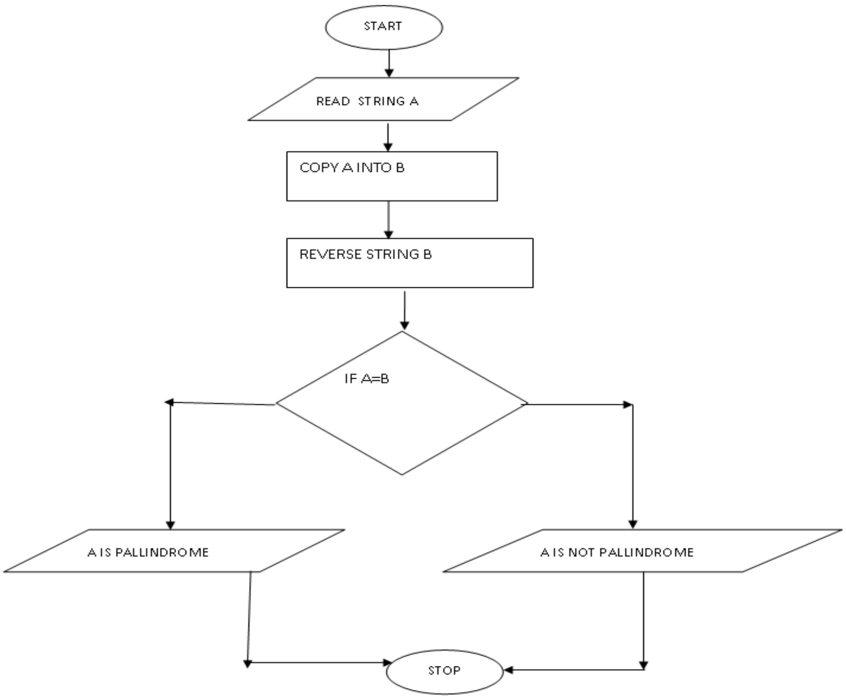
Ex: madam on reversal is also madam

Algorithm:

Step 1: start
Step 2: read string A
Step 3: copy string A into B
Step 4: reverse string B
Step 5: compare A &B
 If A equals B to got step 6
 Else goto step 7
Step 6: print given string A is palindrome
Step 7: print given string is not palindrome
Step 8: stop

Program:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int i,j=0,len=0;
char s1[10],s2[10];
clrscr();
printf("\n Enter any string:");
scanf("%s",s1);
len=strlen(s1);
for(i=len-1;i>=0;i--)
{
s2[j]=s1[i];
j++;
}
s2[j]='\0';
for(i=0;s1[i]!='\0';i++)
{
if(s1[i]==s2[i])
{
continue;
}
else
{
printf("\n Your string is not palindrome");
getch();
exit(0);
}
}
printf("\n Your string is palindrome");
getch();
}
```



Output:

enter any string: liril
Your string is palindrome

10. a) Write a C program to replace a substring with another in a given line of text.

Program:

```
#include <stdio.h>
#include <string.h>
#include<conio.h>
void main()
{
char text[100],word[10],rpwrd[10],str[10][10];
int i=0,j=0,k=0,w,p;
clrscr();
    printf("PLEASE WRITE ANY TEXT.\n");
    printf("GIVE ONLY ONE SPACE AFTER EVERY WORD\n");
    printf("WHEN COMPLETE PRESS  Ctrl-Z  \n");
    gets(text);
    printf("\nENTER WHICH WORD IS TO BE REPLACED\n");
    scanf("%s",word);
    printf("\nENTER BY WHICH WORD THE %s IS TO BE REPLACED\n",word);
    scanf("%s",rpwrd);
    p=strlen(text);
for (k=0; k<p; k++)
 {
  if (text[k]!=' ')
   {
    str[i][j] = text[k];
    j++;
   }
  else
   {
    str[i][j]='\0';
    j=0; i++;
   }
}
str[i][j]='\0';
w=i;

for (i=0; i<=w; i++)
 {
  if(strcmp(str[i],word)==0)
    strcpy(str[i],rpwrd);
    printf("%s ",str[i]);
     }
  getch();
 }
```

Output:

```
PLEASE WRITE ANY TEXT.
GIVE ONLY ONE SPACE AFTER EVERY WORD
WHEN COMPLETE PRESS Ctrl-Z
All work and no play makes Mark a dull boy.
ENTER WHICH WORD IS TO BE REPLACED                  Mark
ENTER BY WHICH WORD THE Mark IS TO BE REPLACED      Tom
All work and no play makes Tom a dull boy.
```

b) Write a C program that reads 15 names each of up to 30 characters, stores them in an array, and uses an array of pointers to display them in ascending (ie. alphabetical) order.

Program:

```
#include <stdio.h>
#include <string.h>
void main()
{
   char name[15][10], tname[15][10], temp[15];
   int i, j, n;

   printf("Enter the value of n \n");
   scanf("%d", &n);
   printf("Enter %d names \n", n);
   for (i = 0; i < n; i++)
   {
      scanf("%s", name[i]);
      strcpy(tname[i], name[i]);
   }
   for (i = 0; i < n - 1 ; i++)
   {
      for (j = i + 1; j < n; j++)
      {
         if (strcmp(name[i], name[j]) > 0)
         {
            strcpy(temp, name[i]);
            strcpy(name[i], name[j]);
            strcpy(name[j], temp);
         }
      }
   }
   printf("\n------------------------------------------\n");
   printf("Input Names \t Sorted names\n");
   printf("------------------------------------------\n");
   for (i = 0; i < n; i++)
   {
      printf("%s \t\t %s \n", tname[i], name[i]);
   }
   printf("------------------------------------------\n");
}
```

Output :

```
Enter the value of n  7
Enter 7 names
heap
stack
queue
object
class
program
project
```

```
----------------------------------------
Input Names    Sorted names
------------------------------------------
heap              class
stack             heap
queue             object
object            program
class             project
program           queue
project           stack
```

11. a) 2's complement of a number is obtained by scanning it from right to left and Complementing all the bits after the first appearance of a 1. Thus 2's Complement of 11100 is 00100. Write a C program to find the 2's complement of a binary number.

Program :

```
#include <stdio.h>
#include<conio.h>
#include<string.h>

void main()
{
 char binnum[50];
 int len, x, y;
 clrscr();
 printf("Enter binary number");
 scanf("%s",binnum);
 len=strlen(binnum);
 x=len-1;
while(binnum[x]!='1')
{
  if(binnum[x]!='0' && binnum[x]!='1')
    {
     printf("Not a binary Number");
    }
  x--;
}
 x=x-1;
while(x>=0)
 {
   if(binnum[x]=='0')
        binnum[x]='1';
   else
        binnum[x]='0';
   x--;
 }
    printf("2's complement of a given binary number is  %s\n",binnum);
getch();
}
```

Output:

Enter binary number: 01101100
2's complement of a given binary number is 10010100

b) Write a C program to convert a positive integer to a roman numeral.
Ex. 11 is converted to XI.

Program:

```
#include<stdio.h>
#include<conio.h>
char roman_Number[10000];
int i=0;

void prefix(char c1, char c2)
{
   roman_Number[i++] = c1;
   roman_Number[i++] = c2;
}

void suffix(char c, int n)
{
   int j;
   for(j=0; j<n; j++)
       roman_Number[i++] = c;
 }

int main()
{
   int j;
   long int number;

   printf("Enter the number you wish to convert to ROMAN Number: ");
   scanf("%d", &number);

 if(number <= 0)
  {
       printf("Invalid number");
       getch();
       return 0;
   }

while(number != 0)
 {

      if(number >= 1000)
      {
          suffix('M', number/1000);
          number = number - (number/1000) * 1000;
      }
       else if(number >=500)
       {
          if(number < (500 + 4 * 100))
           {
             suffix('D', number/500);
             number = number - (number/500) * 500;
           }
          else
```

```
        {
          prefix('C','M');
          number = number - (1000-100);
        }
    }
  else if(number >=100)
    {
      if(number < (100 + 3 * 100))
       {
          suffix('C', number/100);
          number = number - (number/100) * 100;
       }
      else
       {
          prefix('L','D');
          number = number - (500-100);
       }
    }
  else if(number >=50)
  {
      if(number < (50 + 4 * 10))
       {
          suffix('L', number/50);
          number = number - (number/50) * 50;
       }
      else
        {
          prefix('X','C');
          number = number - (100-10);
        }
    }
  else if(number >=10)
   {
      if(number < (10 + 3 * 10))
        {
          suffix('X', number/10);
          number = number - (number/10) * 10;
      }
      else
        {
          prefix('X','L');
          number = number - (50-10);
        }
    }

  else if(number >=5)
    {
      if(number < (5 + 4 * 1))
        {
          suffix('V', number/5);
          number = number - (number/5) * 5;
        }
      else
        {
```

```
                prefix('I','X');
                number = number - (10-1);
              }
          }
        else if(number >=1)
          {
            if(number < 4)
              {
                suffix('I',number/1);
                number = number - (number/1) * 1;
              }
            else
              {
                prefix('I','V');
                number = number - (5-1);
              }
          }
    }

    printf("Equivalent Roman Numeral is : ");
    for(j=0;j<i;j++)
        printf("%c",roman_Number[j]);
    getch();
    return 0;

}
```

Output :

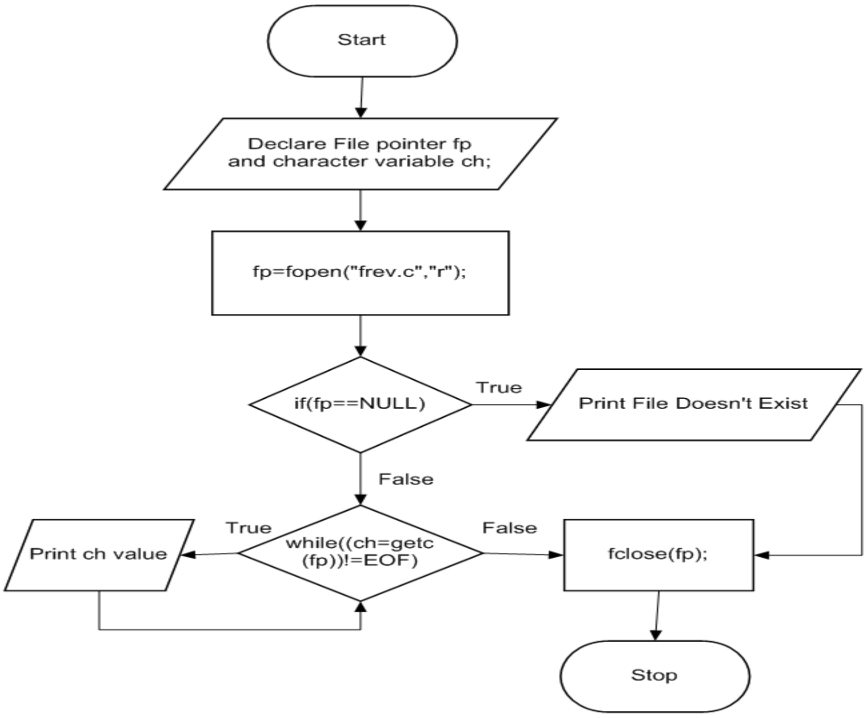
Enter the number: 500
Roman number is be: D

12. a) Write a C program to display the contents of a file to standard output device.

Description: The user will supply the source file name as input and the contents in the source file will be displayed as output.

Algorithm:

Step 1: Start
Step 2: Declare File pointer fp and character variable ch;
Step 3: Open a file in read mode and assign to fp pointer variable
Step 4: Check the condition if(fp==NULL) if true print file doesn't exist else go to step 5
Step 5: Check the condition while(ch=getc(fp))!=EOF) if the condition true go to step 6 until the condition false. If condition false then go to step step7
Step 6: Print ch value
and go to step 5
Step 7: Close the fp file.
Step 8: Stop



Program :

```
#include<stdio.h>
#include<conio.h>
void main()
{
FILE *fp;
char c;
clrscr();
fp=fopen("sample","w");
printf("Enter the text ( Enter * to stop)");
while((c=getchar()!='*')
fputc(c,fp);      //instead of asking the user you can use fputs("india is my country",fp);
fclose(fp);
fp=fopen("sample","r");
printf("contents of a file:\n");
while((c=fgetc(fp))!=EOF)  //getc also works
{
fputc(c,stdout);  //putc also works and we can also use printf("%c",c);
}
fclose(fp);
getch();
}
```

Output:

Enter the text (Enter * to stop) : india is my country *
contents of a file: india is my country

b) Write a C program which copies one file to another, replacing all lowercase characters with their uppercase equivalents.

Program:

```
#include<stdio.h>
#include<conio.h>
void main()
{
FILE *fs,*ft;
char ch;
clrscr();
fs=fopen("source","w");
ft=fopen("target","w");
if(fs==NULL||ft==NULL)
{
printf("\n unable to do the file copy");
exit(0);
}
fputs("C Programing Language",fs);
fcose(fs);
fs=fopen("source","r"); //You have to close the file and open the file again in read mode
while((ch=fgetc(fs))!=EOF)
{
if(ch>='a'&& ch<='z')
ch=ch-32;
fputc(ch,ft);
}
fclose(fs);
fclose(ft);
printf("File successfully Copied and the Target File is:");
ft = fopen("target","r");
while((c=fgetc(ft))!=EOF)
printf("%c",c);
getch();
}
```

Output: File successfully copied and the Target File is: c programming language

13. a) Write a C program to count the number of times a character occurs in a text file. The file name and the character are supplied as command-line arguments.

Program :

```
#include<stdio.h>
#include<conio.h>
void main()
{
int noc=0,now=0,nol=0;
FILE *fw,*fr;
char fname[20],ch;
clrscr();
printf("\n enter the source file name");
gets(fname);
fr=fopen(fname,"r");
if(fr==NULL)
```

```
{
printf("\n error \n");
exit(0);
}
ch=fgetc(fr);
while(ch!=EOF)
{
noc++;
if(ch==' ');
now++;

if(ch=='\n')
{
nol++;
now++;
}
ch=fgetc(fr);
}
fclose(fr);
printf("\n total no of character=%d",noc);
printf("\n total no of words=%d",now);
printf("\n total no of lines=%d",nol);
getch();
}
```

Alternate Program (for understanding):

```
#include<stdio.h>
#include<conio.h>
void main()
        {
int line=1,word=1,ch=0;
char c;
clrscr();
printf("enter text, press Ctrl + z to quit \n");
while((c=getchar())!=EOF)
{
if (c=='\n')
line++;
word++;
else if(c==' ')
word++;
else
ch++;
}
printf("no of lines=%d",line);
printf("\n no of words=%d",word);
printf("\n no of characters=%d",ch);
getch();
}
```

Output:

enter text **india is my country**
all indians are my brothers and sisters

no of lines=2
no of words=11

no of characters=49

14. a) Write a C program to change the nth character (byte) in a text file. Use fseek function.

Program :

```
#include <stdio.h>
#include <conio.h>
void main ()
{
FILE * fp;
char ch;
int n;
clrscr();
fp = fopen("file","w+");
fputs("This is tutorialspoint.com\n", fp);
printf("Enter the position from where you want to change the data");
scanf("%d",&n);
fseek( fp, n, SEEK_SET );
fputs("C Programming Language", fp);
fclose(fp);
fp=fopen("file","r");
while((ch=fgetc(fp))!=EOF)
printf("%c",ch);
fclose(fp);
getch();
}
```

Output: Enter the position from where you want to change the data 8
This is C Programming Language

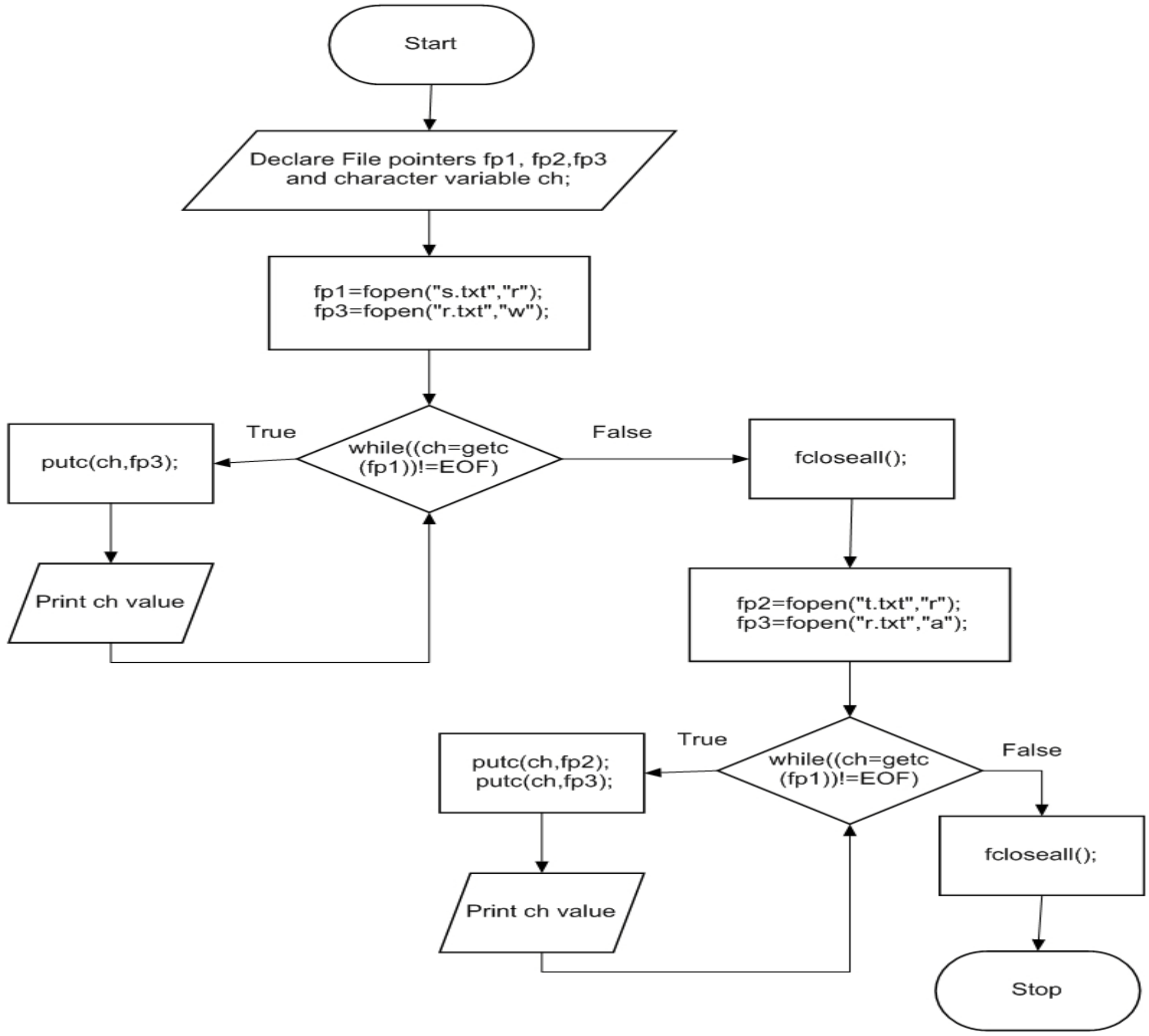
15. a) Write a C program to merge two files into a third file (i.e., the contents of the first file followed by those of the second are put in the third file).

Description:
The user will create three text files and enter contents in two files, the program will merges the contents of these two files in to third file.

Algorithm:

Step 1: Start
Step 2: Declare File pointers fp1, fp2, fp3 and character variable ch;
Step 3: Open a file in read mode and assign to fp1 pointer variable
Step 4: Open a file in write mode and assign to fp3 pointer variable
Step 5: Check the condition while(ch=getc(fp1))!=EOF) if the condition true go to step 6 until the condition false. If condition false then go to step step7
Step 6: Write individual character of a file into fp3 and print the character value and then go to step 5
Step 7: Close all opened files.
Step 8: Open a file in read mode and assign to fp2 pointer variable
Step 9: Open a file in append mode and assign to fp3 pointer variable
Step 10: Check the condition while(ch=getc(fp1))!=EOF) if the condition true go to step 11 until the condition false. If condition false then go to step step12
Step 11: Write individual character of a file into fp2, next write individual characters to fp3 and print the character value and then go to step 10
Step 12: Close all opened files.
Step 13: Stop.

Flowchart:



Program:

```
#include<stdio.h>
#include<conio.h>
void main()
{
FILE *f1,*f2,*f3;
char c;
clrscr();
f1=fopen("sample1.txt","w");
f2=fopen("sample2.txt","w");
fputs("Computer",f1);
```

```
fputs("Programming",f2);
fclose(f1);
fclose(f2);
f1=fopen("sample1.txt","r");
f2=fopen("sample2.txt","r");
f3=fopen("sample3.txt","w");

while((c=fgetc(f1))!=EOF)
fputc(c,f3);
while((c=fgetc(f2))!=EOF)
fputc(c,f3);
printf("Two Files merged into Third File");
fclose(f1);
fclose(f2);
fclose(f3);
getch();
}
```

Output: Two Files merged into Third File

15. b) Define a macro that finds the maximum of two numbers. Write a C program that uses the macro and prints the maximum of two numbers.

Program:

```
#include<stdio.h>
#include<conio.h>

#define Max(A,B)  A>B?A:B

void main()
{
int A,B;
printf("Enter any two numbers");
scanf("%d %d", &A,&B);
printf("Maximum of 2 Numbers is %d", Max(A,B));
getch();
}
```

Output:

Enter any two numbers 10 20
Maximum of 2 Numbers is 20